Mihir Dhanakshirur

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U.S. Citizen | Authorized to work in the U.S. without sponsorship | Google Scholar

Research Interests

Statistical machine learning; learning theory; large language model (LLM) post-training and fine-tuning; probabilistic modeling; reinforcement learning; reasoning and verifiable generation.

Education

University of Michigan, Ann Arbor — Ph.D. in Statistics *Present*
Advisor: Prof. Ambuj Tewari
Relevant coursework: Statistical Machine Learning, Theory of LLMs, Information Theory and Statistics, Probability Theory, Markov Chains, Ergodic Theory and Mixing Times, Stochastic Processes.

Indian Institute of Science, Bengaluru — B.Sc. (Research) in Mathematics *May 2023*
Selected coursework: Reinforcement Learning, Online Learning and Algorithms, Convex Optimization, Causal Inference, Neural Networks and Learning Algorithms.

Research Projects

Entropy Regularization: A Free Correction to Cross-Entropy for Verified Demonstrations *2026*
Accepted: NeurIPS 2026 Workshop on Pre-training to Post-training; also under conference review. [arXiv]

- Developed a learning-theoretic analysis of why cross-entropy (CE) can fail to optimize verifier accuracy; proposed entropy-regularized CE (ER-CE) with PAC-learnability guarantees.
- Fine-tuned Qwen2.5-1.5B and Qwen2.5-7B using LoRA, Hugging Face Transformers/Trainer, PEFT, and custom token-level entropy loss; experiments used NSF ACCESS-supported HPC resources.
- Increased Pass@1 accuracy over CE by **9.04 percentage points** on GSM8K, **4.40** on MBPP, and **5.17** on MATH across math reasoning and code generation.

Combinatorial Characterizations of the Learnability of Conditional Distributions *2026*
Preprint; manuscript under conference review. [arXiv]

- Characterize PAC learnability of conditional-distribution predictors under bounded proper losses, extending classical multiclass learning theory to probabilistic outputs.
- Introduce fat-γ Natarajan and fat-γ Graph dimensions; prove necessary and sufficient learnability criteria and comparison inequalities.
- Prove a coordinate-wise characterization of learnability in the probabilistic concept model; study sample-complexity guarantees.

Research Experience

Imperial College London — Research Intern *Dec. 2022–Aug. 2023*

- Developed continuous Structural Intervention Distance (contSID) with Felix Laumann and Mauricio Barahona, a data-aware approach to comparing causal DAGs via interventional distributions; studied conditional mean embeddings and kernel methods. [arXiv]

Indian Institute of Science — Undergraduate Researcher / Research Collaborator *2023; paper 2026*

- Coauthored work with Gugan Thoppe on robust online distributed linear estimation under adversarial measurements and asynchronous worker updates; established tight finite-time convergence rates and conditions for recovery. [arXiv]

Harvard University — Research Intern *Jun.–Aug. 2023*

- Investigated causal inference methods with Tianle Liu, including kernel-based inverse probability weighting, federated treatment-effect estimation, and doubly robust targeted learning.

Rutgers University, DIMACS REU — Research Student *Jun.–Aug. 2022*

- Studied extensions of the Fortuin–Kasteleyn–Ginibre inequality and extremal properties of monotone Boolean functions; proved special cases.

Technical Skills

Programming and ML: Python, PyTorch, Hugging Face Transformers, Trainer, PEFT, LoRA; LLM supervised fine-tuning, custom loss functions, model evaluation.
Theory and methods: PAC learning, sample complexity, Markov chains and mixing times, information theory, reinforcement learning, causal inference, kernel methods.
Compute: NSF ACCESS-supported high-performance computing (HPC).